

SIGNALS FROM ETI DETECTED - WHAT NEXT?

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Abstract—The question of the formulation of a protocol for activities related to the possible detection of an extraterrestrial civilization is approached on the basis of some factors influencing the degree of impact of the discovery of ETI. Resulting suggestions concerning pre- and post-detection activities are identified.

1. INTRODUCTION

Since 1960, SETI programs realized with differing hardware and under varying assumptions have probed only an infinitesimally small part of the multidimensional search space and were, in fact, doomed to failure - as far as detection of an ETI signal is concerned. They were however, important exercises in formulating the methodology of SETI, and allowing the undertaking of the first large-scale search project: NASA's Microwave Observing Program. That program, to be launched in forthcoming years, will have, for the first time, a considerable chance of finding intelligent signals from outer space.

Possible detection of ETI signals would be, without doubt, one of the most important events in the history of mankind. It may have an enormous impact on the way we think about our place in the Universe, particularly on those geo- and anthropocentrically oriented *weltanschauungen* still existing in some minds.

2. DEGREE OF IMPACT

The degree of impact of ETI detection on terrestrial society will depend, among other things, on three partially interconnected factors:

- A. readiness to accept the fact of ETI detection;
- B. methods of releasing the news of detection;
- C. comprehensibility of the signal's message.

Consideration of such factors may be helpful in the formulation of a protocol for pre- and post-detection activities. The aforementioned three factors are briefly discussed below together with resulting suggestions.

Factor A

The fact of detection of ETI signals can by no means be stated trivially. As shown by Boyce[1], their probable faintness and the possibility of ambiguous interpretation may be the main

obstacles during the identification phase. There exists a more or less widely accepted set of defined characteristics that intelligent signals will presumably have. These are criteria of artificiality, but it would be unreasonable to expect that all of them would be applicable and thereby simplify the identification. In this context the danger of being misled by some criterion should not be overlooked. Tsuriksov proposed, for example[2], that the simultaneous blue- and redshifting of a given spectral feature could serve as a unmistakable sign of the transmission's artificiality; an imitation of Nature's law breaking, since no source can move towards and away from us at the same time. A few years after this proposal, precisely such shifting has been observed in the spectrum of SS443 and explained in a "natural" way[3]. Taking this into consideration, we should be prepared therefore for the reception of a signal which, at best, may be artificial. Before any public announcement of the discovery is made, a great deal of checking and re-checking will be needed, not only at the detection site but also in other observatories. The desire for confirmations as well as the need for 24 hour per day monitoring of the transmitting source will necessarily make the discovery an international affair. Apart from on-site verification procedures adopted by the discovery team, an agreed international protocol should take into account the differences between the observation environment of facilities involved in any confirmation activities, their hardware capabilities, the RFI environment, etc.

It is also of considerable importance that before achieving confirmation, scientists should refrain from making any far-reaching public statements. Well known cases of "artificial" interpretations of CTA-102 radio emission and of U.S. military satellite signals, both prematurely publicized, are quite instructive here. An example of a wiser attitude is the reaction of the British discoverers of pulsars, who kept their findings in secret for a sufficiently long time to allow proof of the natural character of the signals they had received.

Independent reception of the presumably intelligent signal, together with equally independent interpretation of its artificial nature, will strengthen the readiness of the scientific community to accept the fact of ETI's detection. Readiness of the public will depend upon the attitude of opinion-making groups in general and on how and who transmits the news of detection to the society in particular.

Suggestions:

- Further studies on criteria of artificiality for interstellar signals are needed;
- On-site verification protocols as well as international confirmation procedures should be formulated, agreed upon, and disseminated among the SETI community before starting any major search programs;
- It would be advisable for SETI-involved scientists to refrain from making public statements, should a signal of suspected intelligent origin be received.

Factor B

Before considering possible methods of releasing the news of the discovery, two questions should be formulated. First: who will be responsible for making the official announcement about the discovery? Second: how should the announcement be shaped? Detailed answers to both questions are still a matter of consideration and it is to be welcomed that discussions in this area are already underway, involving IISL and the IAA SETI Committee. According to Tarter[4], procedures for handling the detected intelligent signals are being worked on also within the NASA SETI program. It is unknown whether these will take into account any public aspects of the discovery.

Temptation for gaining prestige by showing that the detection is a one-nation achievement cannot be excluded, despite the fact that any signal would be addressed to the entire Earth, if "addressed" is the right word here. For just this reason, one thing seems to be beyond doubt: the official announcement about the discovery of ETI should be shaped and publicized by an international body, preferably one with no political or governmental connections, representing a world-wide network of SETI groups involved in confirmation efforts. Such a body, call it a panel if you prefer, should preferably include not only people from the SETI field, but also outstanding personalities from other areas, just as it was with the group that signed Sagan's SETI petition[5]. It would

be difficult to assess whether a high percentage of Nobel prize laureates in the group influenced the U.S. Congress' decision to start funding the NASA SETI program. However, the program passed suddenly from the "Golden Fleece" category to the class of truly scientific endeavors. Responsibility and credibility - that is what will be needed for any body or panel informing the world about the detection of ETI.

The announcing body should possibly be also the sole source of official information - at least in the first period following discovery. The real societal reactions to the detection of ETI cannot be precisely foreseen; estimates range from hysteria to indifference[6]. An authoritative and prestigious body should therefore be appointed or established, to release information that is not only credible, but also balanced with regard to the signs of societal reactions.

The ability of the mass media to properly interpret the fact of discovery and its possible consequences will be another main factor influencing societal responses to the discovery.

Suggestions:

- Responsibility for public announcement of the discovery of ETI should be placed with an international non-governmental body;
- Societal reactions following the discovery should be continuously studied, influencing the work of the body as an information source;
- The body should include representatives of science journalism.

Factor C

The first ETI signal to be received will probably be a "call" signal, i.e. the kind that is easiest to detect. It might for example contain some simple information about parameters involving "informational" signals that are being transmitted in a different region of the search space. Believing that doing SETI makes sense means that we hope to be clever enough to pick up a "call" signal and to recognize its artificial nature. If the senders are at least equally clever, but not too clever, I would say then we may hope to be able to decipher the data on parameters for the "informational" channel - and even find it. However, difficulties in establishing intercultural communication here on Earth[7] seem to exclude the possibility of comprehending an extraterrestrial message (assuming there will be any message in the signals at all) designed by an intelligence evolved in the course of a separate evolutionary processes, living in a different environment, and quite probably, being far more advanced.

Suggestion:

- Studies on the long-term responses to the discovery of ETI should take into account the likelihood of the reception of artificial but unintelligible signals.

3. CONCLUSIONS

The difficult problem of the formulation of a protocol of activities relating to the detection of an ETI signal can be approached in many ways. The one chosen here, based on consideration of some factors influencing the degree of impact that the discovery of ETI would have, strongly suggests that the significance of such a discovery for humankind should be properly reflected in any adopted protocol, making it an internationally agreed upon document. While observational efforts to detect, verify, and confirm possible ETI signals will be within the competence of scientists, activities following confirmation will certainly call for a broader forum. Since discovery may happen at any time, we should be prepared now to face the problems it may pose.

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